

# Ishan Deshpande

## EDUCATION

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### UT Austin

**M.S. in Electrical and Computer Engineering (Spec. Embedded Systems)**

**December 2026**

**B.S. in Electrical and Computer Engineering**

**May 2025**

3.79 Undergraduate GPA, 3.75 Graduate GPA

**Relevant Coursework:** Embedded Systems Lab, Digital Logic Design, ML/HW Codesign, Comp. Arch., Algorithms, Distributed Systems, Multithreaded Programming, RTOS, VLSI, Graphics, ML Systems, Compilers

## EXPERIENCE

### Apple

**May 2026 - Aug 2026**

*Embedded Intern*

*Cupertino, CA*

- Cut firmware memory footprint by shipping a lockless, interrupt-safe thread-pooling primitive for sharing OS thread stacks
- Quantified memory and performance tradeoffs across firmware subsystems, informing an adoption rollout plan

### Meta Reality Labs

**May 2025 - Aug 2025**

*Codec Avatars Embedded Intern*

*Pittsburgh, PA*

- Accelerated sensor prototyping by building a custom assembler, emulator, and verifier for a programmable I/O state machine on FPGA, reducing development time and enabling flexible sensor integration.
- Improved model fidelity by integrating ultrasonic, radar, and IMU data to capture facial movements.

### Tesla

**May 2024 - Aug 2024**

*Body Controls Firmware Intern*

*Palo Alto, CA*

- Enhanced future firmware reliability by implementing tooling PC-Lint for static code analysis on Renesas microcontroller
- Assessed non-volatile memory usage across all body control modules by introducing CAN logging for EEPROM usage
- Improved vehicle performance by fixing various bugs and implementing features related to sensor calibration and interfacing (mirror heaters, trailer brake sensors, overhead lights, door actuation)

### Garmin

**May 2023 - Aug 2023**

*Flight Control Systems Firmware Intern*

*Olathe, KS*

- Accelerated aircraft control module testing by writing a Python SIL simulation for vertical navigation and altitude tracking
- Optimized flight control module performance by developing HIL unit tests in C within an RTOS environment

### University of Texas at Austin

**Jan 2023 - Present**

*Operating Systems TA*

*Austin, TX*

- Enhance understanding of OS concepts for 70+ students through effective instruction

*Embedded Systems Tutor*

- Elevated students' understanding of embedded systems by teaching key concepts such as interrupts, DAC/ADC, finite state machines, and serial communication, resulting in improved project performance and comprehension

## ACTIVITIES AND PROJECTS

### Longhorn Racing Solar Car

**Sept 2021 - Present**

*Platform Team Lead*

- Improved robustness of on-vehicle firmware updates by developing a UART/CAN bootloader (custom linker script and flash memory driver)

*Controls Team Lead*

- Programmed in an RTOS context for our platform of STM32 microcontrollers (display, motor controller)
- Managed a team of 12 through a traditional software workflow with Git/GitHub

### UT Embedded Systems Lab Design Competition - 3rd Place

**Apr 2024**

*Embedded Software Engineer*

- Collaborated with a team of 4 to develop a GameBoy emulator on a TM4C microcontroller
- Designed a circuit board in KiCAD and worked with protocols such as SPI/QSPI and 16-bit parallel

### UT Real-Time Operating Systems Graduate Project

**Apr 2025**

*Embedded Software Engineer*

- Interfaced with a Wifi-Ethernet bridge chip and wrote MAC, IP, UDP, ICMP, ARP, and DHCP drivers from scratch on a TI microcontroller
- Demonstrated functional ping and netcat implementations on an RTOS

**Skills:** C/C++, KiCAD, Kernel Development, Python, Verilog, Vitis HLS, Application Development for RTOS, Linux, Git, Docker, AWS, Java

**Interests:** Table Tennis, Music Production, Piano